

# Alireza Barzin

AI/ML Engineer | MLOps | AI Application Engineering | ML Platforms | LLMs & RAG

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## SUMMARY

AI/ML Engineer with experience building production-grade ML systems, LLM-powered applications, and agentic AI architectures. Strong background in backend engineering, identity & risk systems, and MLOps pipelines. Skilled in designing scalable services, embedding-based retrieval, tool-calling agents, and secure authentication flows. Earlier experience in quantitative modeling, forecasting, and scenario simulation, providing a strong analytical foundation for ML system design.

## CORE SKILLS

|                             |                                                                    |                                 |                                                                     |
|-----------------------------|--------------------------------------------------------------------|---------------------------------|---------------------------------------------------------------------|
| <b>Machine Learning:</b>    | Scikit-learn, XGBoost, LightGBM, CNNs, RNNs, LSTMs                 | <b>LLMs &amp; RAG:</b>          | LangChain, FAISS, embeddings, OpenAI/Groq APIs, tool-calling agents |
| <b>Backend Engineering:</b> | FastAPI, Flask, REST APIs, microservices                           | <b>Identity &amp; Security:</b> | OAuth/OIDC, JWT, MFA, device trust, risk scoring                    |
| <b>MLOps:</b>               | CI/CD (GitHub Actions, Jenkins), Docker, monitoring, observability | <b>Data Engineering:</b>        | Pandas, NumPy, Spark/PySpark, ETL/ELT, feature engineering          |
| <b>Languages:</b>           | Python, Go, Java, JavaScript, TypeScript                           | <b>Databases:</b>               | Postgres, Redis, pgvector                                           |

## EXPERIENCE

### AI/ML Engineer & Backend Systems Engineer

2025 – Present

Consulting

#### Key Achievements

- Built 5+ end-to-end ML and LLM applications, improving model accuracy, reliability, and deployment readiness across domains.
- Designed agentic AI architectures with tool-calling, multi-step reasoning, and RAG, reducing hallucinations by 40–60%.
- Developed embedding-based retrieval systems (FAISS, pgvector) improving semantic match quality by 30–45%.
- Implemented CI/CD pipelines reducing deployment time by 50% and increasing release reliability.
- Created data pipelines and feature engineering workflows that reduced preprocessing time by 35% and improved model performance by 10–20%.
- Delivered secure authentication flows and backend services supporting thousands of identity events per day.

### Senior Software Engineer — ML Systems & Backend

2022 – 2025

SecureAuth

#### Key Achievements

- Engineered ML-integrated authentication workflows and risk-scoring components, improving fraud detection precision by 25–40%.
- Built backend services for identity, device trust, and anomaly detection, reducing API latency by 30%.
- Developed evaluation and monitoring tools for ML-driven authentication systems, reducing model drift incidents by 35%.
- Implemented CI/CD pipelines (GitHub Actions, Jenkins) improving deployment reliability by 50%.
- Created Python-based validation and automation systems supporting ML-integrated authentication pipelines.
- Authored operational playbooks and deployment guidelines for ML-based authentication services.

### Software Engineer — Backend, Identity, ML Infrastructure

2016 – 2022

Acceptto

#### Key Achievements

- Built backend services and SDK components for identity and behavioral authentication systems across mobile and web platforms.
- Developed Java-based Android SDK modules and contributed to cross-device digital fingerprinting workflows.
- Helped design a secure password vault using React + Java, integrating encryption, session logic, and authentication flows.
- Created Python-based validation and automation systems supporting ML-integrated authentication pipelines.
- Built REST API tools to verify model outputs, risk scores, and system responses under real-world conditions.
- Integrated automated reliability checks into CI pipelines, reducing regression incidents by 30%.

## PROJECTS

### Smart Resume Agent — AI Resume Tailoring Platform

*Python, FastAPI, Next.js, Redis, Postgres, pgvector, Docker*

- Built a four-phase LLM pipeline (keyword extraction → gap analysis → rewrite → QA) improving ATS alignment by 15–30 points.
- Implemented LLM provider abstraction (Gemini, Claude, OpenAI, BYOK) reducing integration overhead by 70%.
- Designed embedding-grounded rewriting using pgvector, eliminating hallucinated experience.
- Created inline editing, version history, and patch-based chat edits, reducing rewrite cycles by 50%.

**Movie Agent Service — Agentic AI for Movie Discovery**

*Python, LangChain, FAISS, Flask, OpenAI/Groq APIs*

- Designed a tool-calling agent architecture with strict separation of reasoning vs. execution.
- Implemented RAG with FAISS, improving retrieval relevance by 40%.
- Built multi-tool orchestration (search, statistics, quiz generation, vision analysis).
- Added validation layers and structured error handling, reducing failure cases by 60%.

**ASAR — Device Identity, Fingerprinting & Risk Engine (Private)**

*Go, TypeScript, Node, OAuth/OIDC, JWT, Argon2id*

- Built device identity + fingerprinting pipeline improving device recognition accuracy by 35%.
- Implemented OIDC discovery, JWKS, and secure internal token issuance.
- Designed anomaly detection engine for behavioral and device-level signals.
- Added Argon2id password storage, account lockout, and hardened login flows.

**Rai — Agentic AI Layer for Identity Workflows (Private)**

*Python, FastAPI, LangChain, FAISS, Kubernetes*

- Built domain-specialist agents with RBAC-scoped tools and fail-closed safety gates.
- Implemented retrieval-augmented generation with structured action flows.
- Added 570+ golden eval cases with CI-gated releases, reducing regressions by >80%.

**Churn Risk Intelligence**

*Python, Pandas, Scikit-learn*

- Built churn prediction models with 15–20% improvement in recall for high-risk segments.
- Designed feature pipelines capturing usage, tenure, and contract behavior.

**Bank Term Deposit Prediction**

*Python, Scikit-learn, Imbalanced-learn*

- Improved minority-class recall by 30–45% using SMOTE + threshold tuning.
- Delivered interpretable models for business decision support.

**Signal Strength Classification**

*Python, Scikit-learn, NumPy*

- Built preprocessing + modeling pipeline improving classification accuracy by 20%.

**EDUCATION**

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**AIML Software Engineering Program**

*IK*

*Machine Learning, Deep Learning, LLMs, MLOps, System Design; with 800/800 capstone project*

**Executive MBA**

*Industrial Management Institute*

*IT Management, Quantitative analysis, forecasting, strategic planning*

*GPA 3.7*

**BSc Economics**

*Allameh Tabataba'i University*

*Statistics, econometrics, modeling*

**Web Development & Design Certificate**

*Portland Community College*

*GPA 3.9*